

```

98.         if move not in ['W', 'A', 'S', 'D']:
99.             print('Invalid direction. Enter one of W, A, S, or D.')
100.            continue
101.
102.            # Check if the player can move in that direction:
103.            if move == 'W' and maze[(playerx, playery - 1)] == EMPTY:
104.                break
105.            elif move == 'S' and maze[(playerx, playery + 1)] == EMPTY:
106.                break
107.            elif move == 'A' and maze[(playerx - 1, playery)] == EMPTY:
108.                break
109.            elif move == 'D' and maze[(playerx + 1, playery)] == EMPTY:
110.                break
111.
112.            print('You cannot move in that direction.')
113.
114.            # Keep moving in this direction until you encounter a branch point.
115.            if move == 'W':
116.                while True:
117.                    playery -= 1
118.                    if (playerx, playery) == (exitx, exity):
119.                        break
120.                    if maze[(playerx, playery - 1)] == WALL:
121.                        break # Break if we've hit a wall.
122.                    if (maze[(playerx - 1, playery)] == EMPTY
123.                        or maze[(playerx + 1, playery)] == EMPTY):
124.                        break # Break if we've reached a branch point.
125.            elif move == 'S':
126.                while True:
127.                    playery += 1
128.                    if (playerx, playery) == (exitx, exity):
129.                        break
130.                    if maze[(playerx, playery + 1)] == WALL:
131.                        break # Break if we've hit a wall.
132.                    if (maze[(playerx - 1, playery)] == EMPTY
133.                        or maze[(playerx + 1, playery)] == EMPTY):
134.                        break # Break if we've reached a branch point.
135.            elif move == 'A':
136.                while True:
137.                    playerx -= 1
138.                    if (playerx, playery) == (exitx, exity):
139.                        break
140.                    if maze[(playerx - 1, playery)] == WALL:
141.                        break # Break if we've hit a wall.
142.                    if (maze[(playerx, playery - 1)] == EMPTY
143.                        or maze[(playerx, playery + 1)] == EMPTY):
144.                        break # Break if we've reached a branch point.
145.            elif move == 'D':
146.                while True:
147.                    playerx += 1
148.                    if (playerx, playery) == (exitx, exity):
149.                        break
150.                    if maze[(playerx + 1, playery)] == WALL:
151.                        break # Break if we've hit a wall.
152.                    if (maze[(playerx, playery - 1)] == EMPTY

```